

Food Delivery Dataset

1. Introduction

The main objective of this project is to predict the delivery time of food orders using machine learning techniques. In modern food delivery systems, accurate delivery time estimation is very important for customer satisfaction and efficient service management.

This project uses historical delivery data to build a predictive model that estimates the delivery time based on various features such as distance, restaurant location, order details, and other factors.

Phase 1

In phase 1 of my project, I selected a dataset related to **Food Delivery Time Prediction** from kaggle and defined methodology that I will apply in late phases of my project.

Phase 2

Data Visualization

In Phase 2, data visualization was performed to better understand the dataset and identify patterns, trends, and relationships between different features.

Various graphs and plots were used to analyze the data, such as:

- **Histograms** to observe the distribution of delivery time
- **Scatter plots** to analyze the relationship between distance and delivery time
- **Bar charts** to compare categorical features
- **Heatmaps** to check correlation between variables

Through visualization, it was observed that:

- Delivery time increases as distance increases
- Some features have stronger influence on the target variable
- There are some outliers present in the dataset

These insights helped in making better decisions for preprocessing and model building.

Data Preprocessing

After understanding the data through visualization, preprocessing was performed to clean and prepare the dataset for machine learning.

The following steps were carried out:

1. Handling Missing Values 2. Removing Duplicates 3. Encoding Categorical Variables 4. Feature Selection 5. Data Formatting 6. Outlier Handling

Phase 3

In Phase 3, feature engineering was performed to enhance the quality of input data and improve the performance of machine learning models. Feature engineering involves creating new features or modifying existing ones so that the model can better understand patterns in the data.

1. Creating New Features

2. Transforming Existing Features

3. Encoding Categorical Variables

4. Feature Selection

5. Handling Skewed Data (if applied)

> Feature engineering significantly improved the model performance by:

- Providing better input representation
- Highlighting important patterns in data
- Reducing noise and irrelevant information

Phase 4

In phase 4, I applied the methodology that I described in phase 1 during submission of my dataset which includes:

- 1.Data Collection
- 2.Data Preprocessing
- 3.Exploratory data analysis
- 4.Model Training

5. Model Evaluation

Phase 5

In this phase, the proposed methodology for food delivery time prediction is compared with recent research work to evaluate its effectiveness. The purpose of this comparison is to analyze how the current approach performs relative to existing methods and to justify its relevance in solving the problem.

Literature Review (Recent Research Work)

To understand the current trends in this domain, five recent research papers (2022 and onwards) related to food delivery time prediction were studied.

Paper 1

Title: Delivery Time Prediction in Food Delivery Services Using Machine Learning

This study applied basic machine learning techniques using limited features such as distance and order details to predict delivery time.

Paper 2

Title: Deep Learning-Based Delivery Time Estimation for Online Food Orders

This paper used deep learning models to improve prediction accuracy. However, the approach is computationally complex.

Paper 3

Title: Urban Logistics and Food Delivery Time Prediction Using Gradient Boosting

This research applied gradient boosting algorithms for prediction but focused on limited feature engineering.

Paper 4

Title: Real-Time Food Delivery Prediction Using Hybrid Machine Learning Models

This study proposed a hybrid approach combining multiple models, increasing complexity.

Paper 5

Title: Predicting Delivery Time with Traffic and Weather Data Using ML

This research incorporated external factors like traffic and weather but used a limited dataset.

Overview of Our Methodology

The methodology used in this project includes:

- Data preprocessing and cleaning
- Data visualization for analysis
- Feature engineering
- Application of machine learning models
- Model evaluation using performance metrics

The approach focuses on building a reliable prediction model using structured data and standard machine learning techniques.

Comparative Analysis

Model Type

Most recent research studies use advanced approaches such as deep learning models or hybrid systems that combine multiple algorithms. These models are designed to improve prediction accuracy but often require complex architectures.

In comparison, this project uses machine learning models, which are simpler and more practical. These models are easier to train and still provide reliable predictions for delivery time.

Model Complexity

Many existing research works involve high complexity, including multiple layers, large datasets, and advanced computations. This makes them difficult to understand and implement.

On the other hand, the methodology used in this project follows a structured and moderate approach. It avoids unnecessary complexity, making the system easier to implement and understand.

Feature Usage

Some research papers focus on limited features such as distance, traffic, or weather conditions. While these features are useful, they may not fully represent all factors affecting delivery time.

In this project, a balanced set of features is used. Different relevant attributes are considered, which helps the model capture better patterns and relationships in the data.

Data Preprocessing

In many studies, only basic preprocessing techniques are applied, which may not be sufficient for improving data quality.

In contrast, this project includes proper data preprocessing steps such as handling missing values, removing duplicates, and cleaning the dataset. This ensures that the data used for training is accurate and consistent.

Implementation and Practicality

Some research approaches are difficult to implement in real-world systems due to their high computational requirements and complexity.

The methodology used in this project is simpler and more efficient. It is easier to implement and can be applied in practical scenarios without requiring advanced resources.

Overall Comparison

Overall, the comparison shows that while existing research focuses on complex and advanced solutions, this project provides a balanced approach. It combines simplicity, efficiency, and effectiveness, making it suitable for practical use while still achieving good prediction performance.

Gap Identification

Based on the literature review, the following observations were made:

- Many research studies use complex models which are difficult to implement
- Some approaches focus on limited features
- Certain models require high computational resources

- Practical implementation can be challenging

Justification of my Work

The methodology used in this project effectively addresses the problem of delivery time prediction by:

- Using a structured and clear approach
- Applying data preprocessing and feature engineering
- Utilizing machine learning models that are efficient and practical
- Providing reliable prediction results

This approach is suitable for real-world applications due to its balance between performance and simplicity.

Conclusion

In this phase, the proposed methodology was compared with recent research work. The comparison shows that while existing studies use advanced and complex models, the current approach provides a simpler and effective solution for delivery time prediction.

References

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